

8. This question is about momentum.

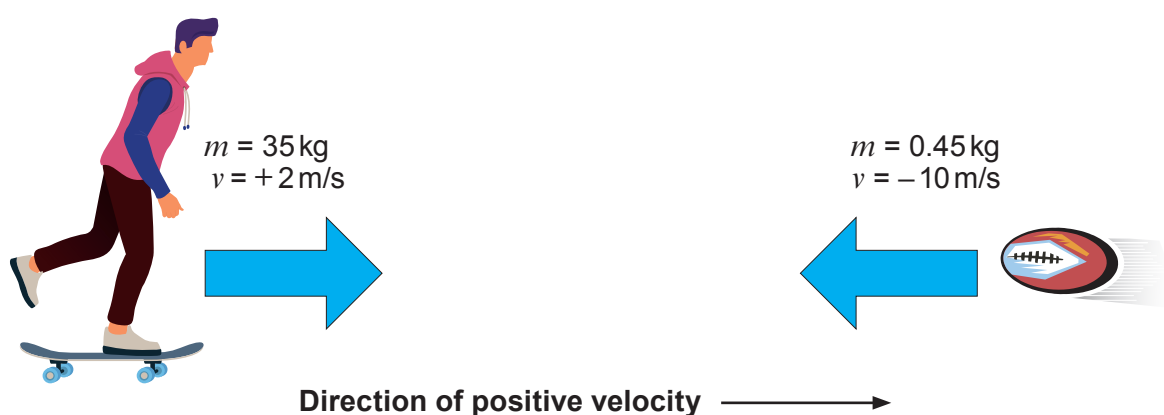
(a) State the principle of conservation of momentum. [2]

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(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit. [4]

Total momentum =

Unit =



- (c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball.

[2]

Velocity = m/s

- (d) Tomos suggests that **kinetic energy** is conserved in this collision. Explain, by using calculations, whether or not you agree. Use an equation from page 2.

[4]

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END OF PAPER

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